Bocconi University

MSc in Data Science and Business Analytics

Modeling Underwriting Syndicates to Predict IPO Performance

Giacomo Bellodi

Master's Thesis

Abstract. This study investigates whether the structure of underwriting syndicates, modeled as a time-evolving hypergraph, carries predictive information about IPO first-day returns. Each syndicate is represented as a weighted hyperedge preserving the collective structure and hierarchical role weights that pairwise co-underwriting graphs discard. Hand-crafted hypergraph features are incorporated into an incremental specification framework and evaluated across XGBoost, Random Forest, Logistic Regression, and a Hypergraph Neural Network on 3,109 IPOs from the US, Canada, and Europe (2013–2026). Firm fundamentals and macroeconomic conditions account for the dominant share of predictive information. Hypergraph features improve multiclass classification for XGBoost but not for other algorithms, and outperform standard pairwise graph features. The HGNN does not match classical models. When the classification signal is translated into a selective subscription strategy, cumulative returns dominate passive and random benchmarks across all market regimes, though the incremental network contribution is not statistically distinguishable at the portfolio level.

2026

Contents

1 Introduction **3**

2 Literature Review **5**

3 Data and Feature Engineering **9**

3.1 Dataset Construction 9

3.2 Target Variable Construction 10

3.3 Hypergraph and Graph Features 11

3.4 Model Feature Specifications 13

3.5 Missing Data Imputation 14

4 Methodologies **15**

4.1 Experimental Design 15

4.2 Classical Machine Learning Models 16

4.3 Hypergraph Neural Network 17

4.3.1 Architecture 18

5 Classification Results **19**

5.1 Task and Horizon Selection 19

5.2 Feature Tier Progression 21

5.3 Ordinal Characterisation of the Best Specifications 24

5.3.1 Full Metric Suite 25

5.3.2 Fold-by-Fold Stability 27

5.4 Summary and Link to the Trading Strategy 28

6 Trading Strategy Evaluation **29**

6.1 Strategy Design and Assumptions 30

6.2 Threshold Sweep and Model Comparison 30

6.3 Performance Against Passive and Random Benchmarks 32

6.4 Interpretation 34

7 Conclusions 35

A Hypergraph and Graph Feature Definitions 40

B HGNN Training Procedure 43

C Full Results 44

1 Introduction

An Initial Public Offering (IPO) occurs when a private company becomes publicly traded by issuing new shares to the public. This process allows investors to buy ownership stakes in the company and enables its shares to be listed and traded on a public stock exchange. The offering is managed by a syndicate of investment banks who underwrite the issuance, conduct the pre-listing roadshow, and set the offer price at which institutional and selected retail investors can subscribe before trading begins. Usually, the offer price is set below the opening price on the public exchange. This phenomenon is referred to as IPO underpricing.

Underpricing represents both a cost to issuing firms and an opportunity for investors capable of identifying which deals will generate the largest first-day returns. The magnitude of the effect is substantial: average first-day returns in the US have ranged from roughly 7% in the 1980s to nearly 65% during the dot-com bubble, and have persisted at economically meaningful levels in more recent periods (J. R. Ritter 1984; J. Ritter and Loughran 2004). A growing body of work has applied machine learning to predict by how much will an IPO underprice, demonstrating that non-linear models trained on firm financials, deal characteristics, and market conditions can outperform traditional regression approaches (Alahmadi 2025; Tsui and Li 2021). Separately, existing literature has established that the structure of underwriting bank syndicate carries information about IPO outcomes, with more central and better-connected lead underwriters associated with distinct pricing patterns (Chuluun 2015; Bajo et al. 2016). These two approaches, however, remain largely disconnected. Machine learning studies incorporate underwriter reputation as a scalar control variable but do not exploit the relational structure of syndicate networks as a source of predictive features. Network studies, conversely, focus on identifying the effects of syndicate characteristics on IPO outcomes rather than integrating network measures into a predictive framework. Moreover, the network representations employed in this literature are exclusively pairwise graphs, which reduce each syndicate to a set of binary edges and discard the group-level and role-specific structure that characterises underwriting relationships.

This study tries to bridge these gaps. Each IPO syndicate is represented as a weighted hyperedge connecting all participating banks simultaneously, giving rise to a time-evolving hypergraph in which banks are nodes and deals are hyperedges. Unlike a standard co-underwriting graph, this representation preserves the collective structure of the syndicate and the hierarchical role weights of its members. From this hypergraph, hand-crafted features are extracted and incorporated into an incremental feature specification framework that isolates their marginal predictive contribution relative to a well-specified no-network baseline. The analysis evaluates XGBoost, Random Forest, and Logistic Regression across nine feature specifications, focusing on multiclass classification at the first-day return horizon (d_0) using an expanding-window cross-validation design with six temporally ordered test folds spanning 2018–2026. Binary classification and longer return horizons are examined as robustness checks. As a methodological complement, a Hypergraph Neural Network (HGNN) that learns bank embeddings directly from the raw syndicate structure via hypergraph convolution is compared against the classical models.

Results from the study show that firm fundamentals and macroeconomic conditions account for the dominant share of out-of-sample predictive information: the largest performance gain occurs before any network features enter the model. Hypergraph network features improve multiclass classification for XGBoost — the algorithm best suited to extracting weak feature interactions — but not for Random Forest or Logistic Regression, and the multi-way hypergraph representation outperforms standard pairwise graph features on the same algorithm. The HGNN does not match the classical models, consistent with sample size constraints on end-to-end representation learning. When translated into a selective IPO subscription strategy, the pre-IPO signal generates cumulative returns that dominate both passive subscription and 10,000 size-matched random portfolios across all tested thresholds and market regimes. However, the addition of network features does not produce a statistically distinguishable improvement in trading performance relative to the no-network baseline, a result confirmed both at the fold level and through deal-level tests on the full set of individual predictions. The classification gains documented are directionally consistent but too small in magnitude to translate into significant portfolio-level differences.

The remainder of the paper is organised as follows. Section 2 reviews the relevant literature on IPO underpricing, underwriter networks, machine learning for IPO prediction, and hypergraph theory. Section 3 describes the dataset, target variable construction, hypergraph and graph feature engineering, and model specifications. Section 4 presents the experimental design and the four modelling approaches. Section 5 reports the classification results, including feature tier progression, ordinal metric analysis, and fold-by-fold stability. Section 6 evaluates the trading strategy and Section 7 concludes and sets the direction for future work.

2 Literature Review

IPO Underpricing The underpricing of initial public offerings (IPOs) remains one of the most enduring puzzles in financial economics. Decades of empirical research have consistently documented that newly listed firms are offered at prices below their first-day closing prices, leaving significant money on the table for issuing firms. J. R. Ritter (1984) was among the first to systematically document this phenomenon, finding average first-day returns of approximately 18% in the US market during hot issue periods. J. Ritter and Loughran (2004) extended this work across multiple decades, showing that average underpricing has not only persisted but intensified over time, rising from roughly 7% in the 1980s to nearly 65% during the dot-com bubble of 1999–2000, before moderating to around 12% in the early 2000s. They attribute this secular shift partly to changes in issuer and underwriter incentives, including the increasing importance of analyst coverage and the practice of spinning—allocating underpriced shares to corporate executives in exchange for future business.

Several theoretical frameworks have been advanced to explain IPO underpricing. Signaling theories, pioneered by Allen and Faulhaber (1989) and Welch (1989), propose that high-quality firms deliberately underprice their IPOs to signal their true value to the market, subsequently recouping this cost through seasoned equity offerings at more favorable prices. Information asymmetry models, most notably the winner's curse framework of Rock (1986), argue that underpricing is necessary to compensate uninformed investors who otherwise face adverse selection against better-informed investors. The certification hypothesis, developed by Booth and Smith (1986), claims that prestigious underwriters certify the quality of the issuing firm, thereby reducing investor uncertainty and the required discount. Benveniste and Spindt (1989) offer a complementary book-building rationale, in which underpricing compensates institutional investors for truthfully revealing their private demand information during the pre-IPO roadshow process. These theoretical advances pushed researchers to further investigate IPO underpricing and how it is driven by underwriter syndicates.

Underwriter Reputation and Certification A central block of the IPO literature has examined how underwriter reputation influences offering outcomes. Carter and Manaster (1990) introduced a widely used ranking system and demonstrated that more reputable underwriters are associated with lower underpricing, consistent with their capacity to certify firm quality and reduce investor uncertainty. Megginson and Weiss (1991) similarly found that venture-capital-backed IPOs underwritten by more prestigious banks exhibit lower underpricing, attributing this to enhanced certification. Ljungqvist and W. J. Wilhelm (2003) revisited these relationships in the context of the dot-com bubble, finding that the traditional negative relationship between reputation and underpricing reversed during this period, suggesting that changes in issuer characteristics and underwriter incentives fundamentally altered market dynamics. The role of syndicate structure has also received considerable attention. Corwin and Schultz (2005) examined IPO underwriting syndicates in detail, finding that the lead underwriter's ability to distribute shares is a critical determinant of IPO success and syndicate composition. They documented that reciprocal co-underwriting relationships between banks are persistent and influence who participates in syndicates. W. Wilhelm and Pichler (2001) offered a theory of syndicate formation, showing that syndicates exist to share marketing costs and improve information production. A critical limitation common to this body of work is its reliance on scalar reputation measures—rankings, market share, or Carter-Manaster scores—that collapse the complex, multi-dimensional relationships between underwriters into a single number. These scalar measures ignore the relational structure of the investment banking industry and cannot capture how the configuration of syndicate members, their roles, and their history of co-underwriting interact to produce IPO outcomes.

Bank Syndicates as Network Structures The application of social network analysis to financial markets has grown substantially over the past two decades. The most directly relevant work on underwriter networks comes from Chuluun (2015) and Bajo et al. (2016). Chuluun (2015) constructed pairwise graph-based network measures (e.g. degree, eigenvector, betweenness centrality) from equity underwriting syndicates between 1970 and 2007, and found that book managers with more central and cohesive networks are associated with higher likelihood of offer price revision and larger price adjustments. Centrality was also positively associated with short-run post-IPO returns, while reciprocity—a measure of the mutual exchange of co-underwriting favors—was positively associated with underpricing, suggesting that well-networked underwriters underprice more to reward their relationship partners. Bajo et al. (2016) extended this framework by analyzing how lead underwriter centrality affects a broader array of IPO outcomes, including investor attention, institutional investor participation, analyst coverage, secondary market liquidity, and long-run returns. They hypothesize and find evidence for two roles of investment banking networks: an information dissemination role, in which central underwriters reach more institutions and attract greater investor attention, and an information extraction role, in which centrality improves the efficiency of book-building. Critically, both of these foundational studies represent underwriter relationships using standard pairwise graphs, where a connection is formed whenever two banks co-underwrite an offering. This approach reduces each syndicate, which may involve five or more banks with distinct hierarchical roles, to a set of binary pairwise edges, discarding the multi-party and role-specific structure of the interaction.

Machine Learning for IPO Prediction The application of machine learning techniques to IPO prediction has grown alongside the broader diffusion of these methods in empirical finance. Early work by Jain and Kini (1994) established the relevance of pre-IPO firm characteristics—profitability, growth, and operating performance—for post-IPO outcomes, providing a foundation for feature engineering in predictive models. Loughran and McDonald (2013) demonstrated the predictive power of textual features extracted from IPO prospectuses, showing that specific linguistic patterns in S-1 filings are associated with subsequent underpricing and long-run performance, opening the door to natural language processing as a source of signal.

More recent work has explicitly adopted machine learning methodologies to predict IPO outcomes. Tsui and Li (2021) apply gradient boosted trees to IPO prospectus readability scores on the Hong Kong Stock Exchange, finding that linguistic features of offering documents carry predictive content for first-day underpricing beyond standard financial controls. Alahmadi (2025) applies a broader ensemble framework — including Random Forest, XGBoost, and Extra Trees — to predict post-listing underperformance, confirming that non-linear methods consistently outperform traditional OLS regression by capturing interaction effects among financial, market, and deal-specific variables. A common finding across this literature is that market sentiment variables — such as recent IPO activity and equity market returns — play an important role in determining short-run IPO returns, consistent with the hot and cold market phenomena documented in the classical literature. Despite these methodological advances, network features have remained largely underexplored in ML-based IPO prediction. Prior studies incorporate underwriter reputation as a scalar control variable but do not construct or evaluate the predictive value of relational network features derived from syndicate structure. This represents a significant gap, given the theoretical and empirical evidence suggesting that underwriter networks carry substantial information about IPO outcomes.

Hypergraphs and Higher-Order Networks The mathematical theory of hypergraphs, formalized by Berge and Minieka (1973) and further developed by Bretto (2013), provides a natural generalization of standard graphs in which edges (hyperedges) can connect any number of nodes rather than being restricted to pairs. This representation is particularly well-suited to capturing group interactions, where the collective structure of the group carries meaningful information. In the social sciences, Battiston et al. (2020) argued forcefully that many real-world social and biological systems are fundamentally higher-order in nature, and that standard graph representations systematically lose information by projecting group interactions onto pairwise edges. They demonstrated that hypergraph and simplicial complex representations can reveal dynamics and structural properties that are invisible to conventional network analysis, including the role of group cohesion and collective behavior in information diffusion and social influence. Applications of higher-order network representations in financial settings remain an emerging area, and the projection of syndicate interactions onto pairwise graphs discards precisely the multi-party, role-specific structure that characterises underwriting relationships. This study addresses that gap directly, incorporating the hypergraph structure of bank syndicates as a source of predictive signal for IPO first-day returns.

3 Data and Feature Engineering

3.1 Dataset Construction

The dataset is sourced from FactSet Research Systems Inc. (2026) and covers IPOs and equity offerings in the United States, Canada, and Europe between 2013 and 2026. The geographic and temporal scope is chosen to maximise sample size while maintaining data quality and coverage of recent market cycles, yielding an initial universe of approximately 7,000 deals. Three listing types are retained: Underwritten Offerings, Offers for Subscription, and Offers for Sale/Shareholder Sale. Listing types that deviate from the standard bookbuilt issuance structure are excluded, as they differ materially in their pricing mechanism and syndicate composition.[1] Deals with missing core transaction data are removed. To limit microstructure noise associated with highly speculative issues, the sample is restricted to IPOs with an offer price of $1.00 or above, following the idea of cutoff criterion of Bradley and Jordan (2002). The final sample comprises 3,109 deals.

For each deal, features are extracted across four categories: deal characteristics (offer

[1]Excluded types: Placing/Private Placement, Up-listing, Mutual to Stock Conversion, Mutual Holding Company to Stock Conversion, Mutual to Mutual Holding Company Conversion.

price, deal size, shares offered, listing exchange, issuer sector and country); issuer financials (revenue, EBITDA, net income, total assets, and cash holdings); macroeconomic controls (S&P 500 return, VIX level, 10-year Treasury yield, and lagged IPO count); and syndicate network features derived from a hypergraph representation of underwriting relationships, described in Section 3.3. Closing prices are collected at the first trading day (d_0), and at approximately one week (d_4), two weeks (d_9), and one month (d_{20}) post-listing, from which holding-period returns are computed.

3.2 Target Variable Construction

The primary outcome variable is the return of each deal relative to its offer price at horizon h:

$$r_{i,h} = \frac{P_{i,h} - P_{i,\text{offer}}}{P_{i,\text{offer}}} \tag{1}$$

where $P_{i,h}$ is the closing price of deal i at horizon h after the first trading date. Three horizons are constructed: the first-day close (d_0), the close after nine trading days (d_9, approximately two calendar weeks), and the close after twenty trading days (d_{20}, approximately one calendar month).

Raw returns are winsorised at the 1st and 99th percentiles within each training fold to limit the influence of extreme observations on model training. Winsorisation boundaries are computed exclusively from training data and applied to test folds without refitting, preserving the integrity of the out-of-sample evaluation.

Two parallel classification targets are constructed from these returns. The binary target partitions deals by the sign of the return: Class 0 (Negative, $r_{i,h} \leq 0$) and Class 1 (Positive, $r_{i,h} > 0$). At d_0, 64.9% of deals are Positive, reflecting the well-documented first-day underpricing tendency; this fraction declines moderately to 62.9% at d_9 and 61.7% at d_{20} as post-listing drift erodes some initial gains. The resulting approximately 2:1 class imbalance is moderate; all models are evaluated on AUC rather than accuracy to remain robust to it.

The multiclass target introduces a third class to distinguish economically distinct return regimes:

- **Class 0 — Negative**: $r_{i,h} \leq 0\%$
- **Class 1 — Moderate**: $0\% < r_{i,h} \leq 15\%$
- **Class 2 — Hot**: $r_{i,h} > 15\%$

The 15% threshold is chosen to separate economically distinct return regimes while producing approximately balanced class proportions at the primary horizon: at d0 the three classes contain 35.1%, 35.3%, and 29.6% of deals respectively. At d_9 and d_{20} the Moderate class shrinks (to 29.5% and 27.3% respectively) as short-term drift pushes deals toward the tails, consistent with post-IPO price support and momentum documented in the aftermarket literature.

3.3 Hypergraph and Graph Features

Hypergraph-Motivated Features (HG) [2]

Each deal is associated with a set of participating banks and their syndicate roles: Global Coordinator, Bookrunner, Co-Manager, and so forth. Rather than reducing these multi-way relationships to pairwise links, each IPO syndicate is represented as a hyperedge connecting all participating banks simultaneously, giving rise to a time-evolving hypergraph in which banks are nodes and deals are hyperedges. Each bank's participation is weighted according to its syndicate role, following a hierarchical scheme in which senior positions receive higher values. The existence of a clear pecking order within syndicates is well-documented (Corwin and Schultz 2005; Chemmanur and Krishnan 2012); the specific numerical weights assigned to each tier (Table 8 in Appendix A) are a design choice of this study, calibrated to preserve the ordinal ranking. Formal sensitivity to alternative schemes is left for future work.

Three families of features are extracted from this representation, computed from a rolling three-year window of historical syndicate activity strictly prior to each deal's first trading date, ensuring zero look-ahead leakage. The families correspond to the three network blocks that enter the model specifications incrementally.

[2] Formal definitions of all features and the role-weighting scheme are provided in Appendix A.

The first family (HG_R) captures the internal composition of the syndicate. The distribution of role weights across participating banks gives rise to measures of intensity and diversity, measured by the Shannon entropy of the role weight distribution.

The second family (HG_P) measures the historical activity and experience of the banks participating in the focal deal: the sum, mean, and maximum of weighted bank degree in the rolling window capture the aggregate experience and activity intensity of participating banks, and the number of active banks in the snapshot controls for market-wide syndicate density at the time of the deal.

The third family (HG_A) extends the characterisation to pairwise collaboration history and leadership structure. Mean and maximum pairwise co-underwriting counts among syndicate members, together with the fraction of pairs with at least one prior co-deal, measure how familiar the syndicate members are with each other. The weighted degree of the lead bank and the dominance ratio, capturing the concentration of seniority at the top of the hierarchy, complete this block.

Not all of these features require the hypergraph data structure to compute: HG_R describes properties of a single syndicate, HG_A's pairwise co-counts could be derived from a standard graph, and only HG_P's role-weighted degree exploits the incidence matrix directly. The hypergraph framework serves primarily as a conceptual lens that motivates role-weighted, group-aware feature design — an orientation that a pairwise graph perspective does not naturally foreground. Where the hypergraph is computationally irreplaceable is in the HGNN (Section 4.3), whose convolution operator is defined directly on the role-weighted incidence matrix.

Standard Graph Features (GR) As a benchmark, a parallel set of features is constructed from the standard co-underwriting graph, in which two banks are connected whenever they have co-underwritten at least one deal in the rolling three-year window. Unlike the features above, this block relies on classical graph-theoretic measures — centrality and clustering — that operate on binary pairwise topology, without access to role weights or group-level syndicate structure. Eight statistics are extracted for each deal:

mean and maximum clustering coefficients, capturing local network cohesion; mean and maximum PageRank centrality, measuring each bank's importance within the broader network; mean and maximum betweenness centrality, reflecting the extent to which syndicate banks bridge otherwise disconnected parts of the network; and the total number of edges and density of the snapshot graph, controlling for overall market connectivity. Their inclusion in M6 — where they substitute for the hypergraph blocks — tests whether graph-theoretic topology measures capture the same information as the role-weighted, group-aware features motivated by the hypergraph framework.

3.4 Model Feature Specifications

To isolate the incremental contribution of syndicate network information, the analysis is structured around nine feature specifications M0–M7. Each adds a distinct set of covariates to its predecessor, enabling a clean attribution of performance changes to the marginal feature group introduced. All specifications are held fixed across all algorithms, classification tasks, and prediction horizons.

Six feature groups enter the specifications. *Market controls* (11 features) capture the macroeconomic environment at listing: equity returns, volatility, interest rates, and IPO market activity. *Deal characteristics* (6 features) include offer size, offer price, shares offered, and syndicate composition counts. *Core financials* (14 features) cover size, profitability, leverage, and liquidity, together with missing-value indicators for the four most frequently absent variables; the *extended financial* block (23 features) augments these with additional accounting ratios. Financial features were selected on theoretical grounds and prior IPO literature (J. R. Ritter and I. Welch 2002), without reference to held-out test performance. *Sector and exchange dummies* (26 features) control for industry heterogeneity and listing-venue fixed effects. The four *network blocks*: hypergraph role ($\mathrm{HG_R}$, 5), bank-level activity ($\mathrm{HG_P}$, 4), additional hypergraph ($\mathrm{HG_A}$, 5), and standard graph (GR, 8).

Table 1 summarises the composition of all specifications. M0 is a naive benchmark using only market controls and categorical dummies; M1 adds deal characteristics; M2 adds core financials, forming the primary no-network baseline. M2 is also the tabular feature set used by the HGNN architecture (Section 4.3), ensuring parity between classical and neural network inputs. The network specifications build incrementally on M2: M3 introduces hypergraph role features in isolation; M4 adds bank-level activity; M4b completes the hypergraph set by appending the additional features; M5 is the saturated specification combining the extended financial block with the full hypergraph set. M6 substitutes standard graph topology measures for the hypergraph block, testing whether the multi-way representation captures information beyond a conventional co-underwriting graph. M7 combines all feature groups as a kitchen-sink robustness check.

Table 1: Feature composition of the nine model specifications. Each specification adds a distinct set of covariates to its predecessor, enabling attribution of performance changes to the marginal feature group introduced. ✓ = included; blank = excluded. Mkt: market controls (11); Deal: deal characteristics (6); Fin_C: core financials (14); Fin_E: extended financials (23); HG_R: hypergraph role features (5); HG_P: hypergraph bank-level activity features (4); HG_A: additional pairwise and hypergraph features (5); GR: graph features (8); S+E: sector and exchange dummies (26); p: total feature count. †Primary no-network baseline. ‡Central network specifications.

	Feature blocks									
Spec.	Mkt	Deal	Fin_C	Fin_E	HG_R	HG_P	HG_A	GR	S+E	p
M0	✓								✓	37
M1	✓	✓							✓	43
M2†	✓	✓	✓						✓	57
M3‡	✓	✓	✓		✓				✓	62
M4‡	✓	✓	✓		✓	✓			✓	66
M4b‡	✓	✓	✓		✓	✓	✓		✓	71
M5‡	✓	✓		✓	✓	✓	✓		✓	80
M6	✓	✓	✓					✓	✓	65
M7	✓	✓		✓	✓	✓	✓	✓	✓	88

3.5 Missing Data Imputation

Financial data for each issuer were collected as of the fiscal year preceding the IPO. Given the private nature of companies prior to listing, FactSet coverage is incomplete. Missing value rates are 27.4% for revenue, 18.6% for EBITDA, 17.9% for cash and short-term investments, 17.2% for free cash flow, and 14.2% for net income; total assets and liabilities are missing for 14.3% of deals each. Issuer age is missing for 6.0% of observations. Missing values are imputed using the within-fold training median, and binary indicator variables are added for each imputed observation, allowing the model to distinguish between a reported zero and an absent value. Imputation boundaries are never computed from test data.

4 Methodologies

This study evaluates two families of predictive models. The first comprises XGBoost, Random Forest, and Logistic Regression.These models consume hand-crafted tabular features, including the network statistics described in Section 3.3. The second is a Hypergraph Neural Network (HGNN), which learns bank embeddings directly from the raw syndicate structure via hypergraph convolution, and combines them with the same tabular features at the deal level. The two families therefore exploit the same underlying relational data in fundamentally different ways: the classical models receive a fixed, interpretable summary of the hypergraph as additional columns, while the HGNN discovers latent node representations end-to-end. Comparing their out-of-sample performance provides both a practical benchmark and a methodological assessment of whether a learned network representation adds value over hand-engineered alternatives on a dataset of this size. The economic significance of the predictive signal is evaluated through a selective IPO subscription strategy described in Section 6.

4.1 Experimental Design

To avoid look-ahead bias, the study adopts an expanding-window cross-validation scheme in which the training set grows monotonically with each fold and the test set always lies strictly in the future relative to the entire training period.

The dataset spans IPO completions from 2014 to 2026.[3] Six folds are defined with test periods covering 2018 through 2026, as summarised in Table 2. All preprocessing steps that depend on the data distribution — including median imputation of missing financial values and, for Logistic Regression, feature standardisation — are fitted exclusively on

[3]Deals from 2013 are used to initialise the hypergraph rolling window but are excluded from training and evaluation, as insufficient historical syndicate data precedes them.

the training portion of each fold and applied to the test set without refitting.

Table 2: Expanding-window cross-validation folds. The training set grows cumulatively with each fold and the test set always lies strictly in the future relative to the entire training period. n_{train} and n_{test} represent the number of IPO deals included in each training and test fold.

Fold	Training period	Test period	n_{train}	n_{test}
Fold 1	$\leq$ 2017	2018	1,014	312
Fold 2	$\leq$ 2018	2019	1,326	222
Fold 3	$\leq$ 2019	2020	1,548	293
Fold 4	$\leq$ 2020	2021	1,841	689
Fold 5	$\leq$ 2021	2022–23	2,530	254
Fold 6	$\leq$ 2023	2024–26	2,784	322

For each horizon, both classification tasks are considered. The full experimental grid comprises 9 model specifications $\times$ 3 algorithms $\times$ 2 tasks $\times$ 3 horizons $\times$ 6 folds = 972 classification experiments for the classical methods, plus 36 fits for the HGNN. Area under the ROC curve (AUC) is the primary evaluation metric, using one-vs-rest macro-averaging for the multiclass task to prevent majority-class dominance.

To maintain a focused empirical narrative, a primary evaluation setting is used constantly throughout Section 5: multiclass classification at the first-day horizon (d_0). The multiclass task identifies Hot IPOs and captures economically meaningful heterogeneity that the binary task loses by collapsing Moderate and Hot outcomes into a single Positive class. Additional empirical justification is proposed in Section 5.1, where performance is compared across tasks and horizons for a representative set of specifications.

4.2 Classical Machine Learning Models

Three well-established classifiers are trained on the tabular feature sets defined in Section 3.4.

XGBoost (Chen and Guestrin 2016) is a gradient-boosted tree ensemble that minimises a regularised objective via second-order approximation. It is the primary model of this study, selected for its strong empirical performance on structured financial data, its native handling of missing values, and its built-in ℓ_1/ℓ_2 regularisation. Class imbalance is addressed through the `scale_pos_weight` parameter for the binary task and inverse-frequency sample weights for the multiclass task. Hyperparameters — including tree depth, learning rate, subsampling rates, and regularisation coefficients — are tuned per fold via 15-iteration randomised search with 3-fold cross-validation on the training set, scoring on AUC-OVR for the multiclass task and standard AUC for the binary task. The final model is retrained on the full training fold using the selected hyperparameters.

Random Forest (Breiman 2001) is a bagged ensemble of decision trees with feature subsampling at each split. It provides a natural benchmark for XGBoost and is less sensitive to hyperparameter choices, making it a robust alternative. Class balance is handled through the `class_weight` argument and hyperparameters are tuned via 10-iteration randomised search.

Logistic Regression serves as a linear baseline that imposes no interaction structure. Features are standardised to zero mean and unit variance within each training fold before fitting. The regularisation strength C is selected from a five-point grid $\{0.001, 0.01, 0.1, 1, 10\}$ via 3-fold cross-validation. Convergence is ensured by setting a maximum of 2,000 iterations, which is necessary for the higher-dimensional specifications M5 and M7.

4.3 Hypergraph Neural Network

The HGNN provides an end-to-end alternative to the hand-crafted hypergraph features consumed by the classical models. Rather than summarising the syndicate structure into a fixed feature vector, the HGNN learns a continuous embedding for each underwriting bank by propagating information through the full hypergraph of bank-deal relationships, and aggregates these embeddings at the deal level to produce an IPO-level representation for classification. At each training fold, the incidence matrix $\mathbf{H} \in \mathbb{R}^{N_B \times N_D}$ is constructed from all deals in the training window, where N_B is the number of active banks and N_D is the number of training deals. Each non-zero entry H_{bd} is set to the role weight of bank b in deal d, encoding the hierarchical importance of each bank's participation. Banks not observed in the training window are excluded from test-time embedding, with their deals receiving a zero bank embedding vector. Each bank b is initialised with an

18-dimensional feature vector capturing its historical activity profile as of the training cutoff: log deal count and volume, average deal size, recency of last deal, role mix across syndicate positions, past IPO return statistics (mean return, win rate, volatility), co-underwriting network statistics (number of co-underwriters, total co-appearances, average repeat collaboration), and temporal activity patterns (active years, deals per year, recent-year intensity). All bank features are computed exclusively from training-window data.

4.3.1 Architecture

The HGNN encoder follows the normalised hypergraph convolution of Feng et al. (2019). For input bank feature matrix $\mathbf{X} \in \mathbb{R}^{N_B \times d}$ and incidence matrix $\mathbf{H}$, each convolution layer computes:

$$\mathbf{X}' = \mathbf{D}_v^{-1/2}\,\mathbf{H}\,\mathbf{D}_e^{-1}\,\mathbf{H}^\top\,\mathbf{D}_v^{-1/2}\,\mathbf{X}\,\mathbf{W} \tag{2}$$

where $\mathbf{D}_v$ and $\mathbf{D}_e$ are the diagonal degree matrices of vertices and hyperedges respectively, and $\mathbf{W}$ is a learnable weight matrix. Two such layers with hidden dimensions $[128, 64]$ produce a 64-dimensional embedding for each bank. Batch normalisation and dropout ($p = 0.20$) are applied between layers.

At the deal level, bank embeddings are aggregated via a role-weight-normalised sum:

$$\mathbf{e}_{\mathrm{IPO}} = \sum_{b \in \mathcal{S}_d} \frac{w_b}{\sum_{b' \in \mathcal{S}_d} w_{b'}}\,\mathbf{z}_b \tag{3}$$

where $\mathcal{S}_d$ is the syndicate of deal d, w_b is the role weight of bank b, and $\mathbf{z}_b \in \mathbb{R}^{64}$ is its learned embedding. The deal embedding is concatenated with the 57-dimensional M2 tabular feature vector and passed to a two-layer MLP classification head with hidden dimensions $[128, 64]$. Full training details are provided in Appendix B.

5 Classification Results

This section presents the out-of-sample classification results obtained from the expanding-window evaluation framework described in Section 4.1. The analysis proceeds in three stages. Section 5.1 selects the primary evaluation setting by comparing performance across classification tasks and return horizons. Section 5.2 examines how predictive performance evolves across the feature tier hierarchy (M0 through M7), isolating the contribution of financial, deal, market, and hypergraph features to out-of-sample discrimination. Finally, Section 5.3 provides a deeper characterisation of the best-performing specifications through ordinal and boundary-level metrics, establishing the link between classification performance and the trading strategy of Chapter 6.

Throughout this chapter, unless otherwise stated, results refer to XGBoost as the primary algorithm.

5.1 Task and Horizon Selection

Before presenting the main comparative results, I establish empirically which classification task and return horizon yield the most informative evaluation setting. As motivated in Section 4.1, the choice of multiclass classification at the d_0 horizon is grounded in the structure of the investor's decision problem: the question of whether to subscribe to an IPO at the offer price is inherently a first-day question, and the relevant distinction for a selective strategy is between Hot deals (first-day return $> 15\%$), Moderate deals (0–15%), and Negative outcomes ($< 0\%$), not merely between positive and negative returns. We now show that this choice is also supported by the out-of-sample evidence.

Table 3 reports binary AUC and multiclass AUC-Macro (one-vs-rest, macro-averaged) at the d_0 horizon for five key representative specifications: M2 (the no-network baseline), M3 (first family of hypergraph features), M5 (the full hypergraph specification), M7 (kitchen-sink), and HGNN (learned embedding). Results are averaged across the six expanding-window test folds and, for the classical algorithms, across XGBoost, Random Forest, and Logistic Regression.

Table 3: Binary AUC and Multiclass AUC-Macro at the first-day horizon (d_0), averaged across six expanding-window test folds (2018–2026). Classical algorithm columns show the mean across XGBoost, Random Forest, and Logistic Regression. Δ denotes the gain of the multiclass formulation over binary. M2 is the no-network baseline (market controls, deal characteristics, core financials); M3 adds hypergraph role features; M5 is the saturated specification combining extended financials with all hypergraph features; M7 includes all feature groups. HGNN learns bank embeddings directly from the hypergraph and does not consume hand-crafted feature tiers; it is reported separately. Cells marked — indicate that the combination is not applicable: classical algorithms are not evaluated on the HGNN architecture, and the HGNN does not operate on hand-crafted feature specifications.

	Classical algorithms (mean)		**HGNN**	
Model	Binary AUC	MC AUC-Macro	Binary AUC	MC AUC-Macro
M2	0.6437	0.6880	—	—
M3	0.6455	0.6841	—	—
M5	0.6575	0.6879	—	—
M7	0.6502	0.6872	—	—
HGNN	—	—	0.5942	0.6135
Mean Δ	+0.038		+0.019	

Two findings emerge from Table 3. First, the multiclass formulation consistently outperforms binary classification across all specifications and all algorithms, with a mean advantage of 3.8 percentage points for the classical models and 1.9 points for HGNN. This gap reflects a fundamental difference in what the two tasks ask the model to learn: the binary task conflates Moderate and Hot outcomes into a single Positive class, discarding the very information that distinguishes an ordinary first-day gain from a large underpricing event. The multiclass formulation preserves this distinction, allowing the model to concentrate probability mass on the Hot class — the signal that drives greater underprice exposure.

Second, the advantage of multiclass over binary is robust to model specification: the gap ranges from +0.030 (M5 classical) to +0.044 (M2 classical), with no specification where binary classification matches or exceeds multiclass. This consistency across five qualitatively different specifications rules out the possibility that the multiclass advantage is an artifact of a particular feature set.

Table 4 further shows that the d_0 horizon yields the highest AUC-Macro across the four principal model tiers, with performance declining monotonically as the horizon lengthens to d_9 (approximately two weeks) and d_{20} (approximately one month). This is consistent with the interpretation that offer-price underpricing is priced in on the first trading day, while subsequent returns are increasingly driven by market-wide and firm-specific factors that are not captured by pre-IPO observable characteristics.

Table 4: Multiclass AUC-Macro OVR by return horizon for XGBoost, averaged across six expanding-window test folds (2018–2026). d_0: first-day close; d_9: close after nine trading days (approximately two weeks); d_{20}: close after twenty trading days (approximately one month). M2 is the no-network baseline; M3 adds hypergraph role features; M5 combines extended financials with all hypergraph features; M7 includes all feature groups.

Model	d_0	d_9	d_{20}
M2	0.6955	0.6914	0.6789
M3	0.6923	0.6929	0.6839
M5	0.6980	0.6891	0.6785
M7	0.6964	0.6872	0.6803

A partial exception is M3 at d_9 (0.6929), which marginally exceeds M3 at d_0 (0.6923). However, for the purposes of the main analysis , the d_0 horizon is the relevant decision point: an investor decides whether to subscribe before trading begins and realizes the first-day return at the close of the first trading day.

Based on Table 3 and Table 4, the remainder of this chapter focuses on multiclass classification at the d_0 horizon. Appendix C reports the complete classification results for all algorithms, specifications, and horizons.

5.2 Feature Tier Progression

Figure 1 traces the evolution of out-of-sample AUC-Macro across the nine model tiers (M0 through M7) for each of the three classical algorithms, evaluated at the d_0 horizon under the multiclass setting. Each point represents the mean AUC-Macro across the six expanding-window test folds. The HGNN is shown as a horizontal reference line ($\mu = 0.609$, dotted) since it does not operate on a hand-crafted feature tier.

The dominant lift is M0 to M2. The most substantial performance gain in the feature hierarchy occurs before any network information is introduced. Moving from the structural baseline M0 containing only market features and sector and exchange dummies

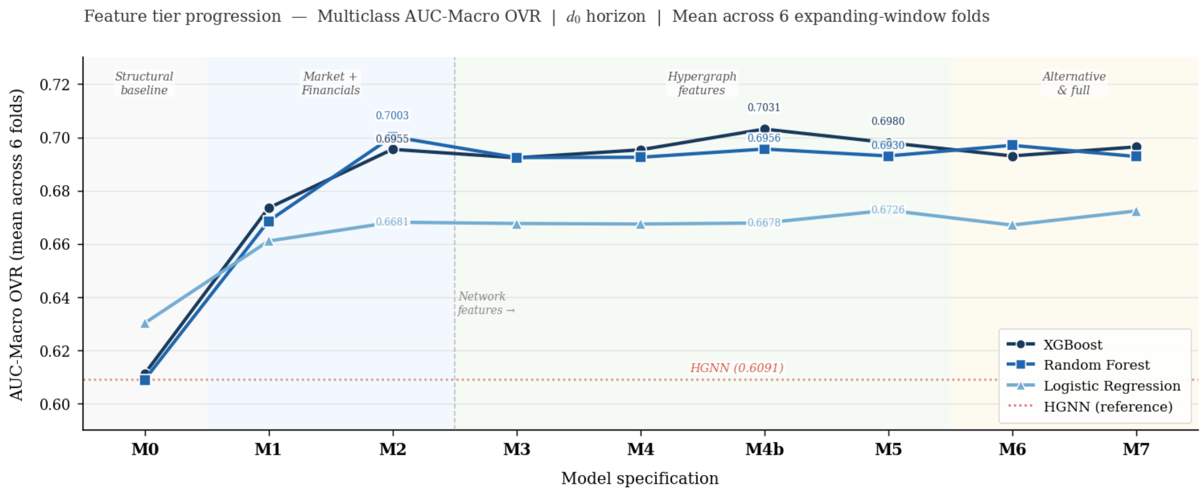

Figure 1: Multiclass AUC-Macro OVR at the first-day horizon (d_0), plotted across feature tiers M0–M7 and M4b for XGBoost, Random Forest, and Logistic Regression. Each point is the mean over six expanding-window test folds (2018–2026). The dotted red line marks the HGNN mean AUC-Macro (0.609) for reference. Background shading groups tiers into four conceptual regions: structural baseline (M0: market controls and sector/exchange dummies only), market and financial features (M1: adds deal characteristics; M2: adds core financials, serving as the no-network baseline), hypergraph network specifications (M3: adds role features; M4: adds bank-level activity; M4b: adds pairwise and hierarchical features; M5: replaces core with extended financials), and alternative network and kitchen-sink specifications (M6: substitutes standard graph centrality for the hypergraph blocks; M7: combines all feature groups). AUC-Macro is computed via one-vs-rest macro-averaging across the three classes (Negative, Moderate, Hot).

to the full no-network specification M2, which adds market conditions, deal characteristics, and core financial variables, yields an improvement of +8.4 pp for XGBoost (0.611 to 0.696), +9.1 pp for Random Forest (0.609 to 0.700), and +3.8 pp for Logistic Regression (0.630 to 0.668). This confirms that firm fundamentals and market conditions carry the dominant fraction of predictive information about first-day returns, consistent with the established literature on IPO pricing determinants (J. R. Ritter and I. Welch 2002; J. Ritter and Loughran 2004). The M1 intermediate tier, which adds market and macro controls to the sector and exchange dummies, accounts for roughly two-thirds of this lift, underscoring the importance of the macroeconomic environment in which an IPO occurs.

Network features are informative for XGBoost, not for Random Forest. The response to network feature additions from M2 onwards is heterogeneous across algorithms. XGBoost improves monotonically from M2 through M4b (+0.8 pp, reaching 0.703) before settling at M5 (0.698), with all network-augmented tiers above the M2 baseline. By contrast, Random Forest declines at every network-augmented tier relative to M2: M3 falls to 0.692 (−0.8 pp), and the degradation persists through M5 (0.693, −0.7 pp) and M7 (0.693, −0.7 pp). Logistic Regression is essentially flat across the network tiers (M2 through M7 span a range of just 0.5 pp), with a modest recovery at M5 and M7 attributable to the extended financial block rather than the network features themselves.

This pattern is explained by the nature of these algorithms. Random Forest builds trees by randomly subsampling both observations and features at each split, which means the marginal network features — relatively few in number and carrying a weaker signal than the dominant financial variables — are diluted by the random subspace procedure and rarely selected at the top of the tree. XGBoost, by contrast, learns feature interactions through sequential residual fitting and explicitly regularises tree complexity, allowing it to identify and amplify weak but consistent signals from the syndicate structure. Therefore, as XGBoost is more capable of learning feature interactions to be extracted, the study focuses on this algorithm for the remainder of the analysis and for the trading strategy of Section 6.

M4b and M5 are the strongest specifications for XGBoost. Within the network region, M4b (0.703) achieves the highest point estimate for XGBoost, followed by M5 (0.698) and M4 (0.695). The M4b specification — which pairs the core financial block with the full hypergraph feature set — outperforms M5 at this horizon despite M5 incorporating a richer financial block. This suggests that, at the first-day horizon, the additional financial variables in the extended block (M5 relative to M4b) do not add predictive value beyond what is already captured by the core financials, and may introduce mild noise through overfitting to the training window. M5 recovers its advantage in the binary task and at longer horizons (see Appendix C,Table 9), consistent with the extended financials carrying signal over a longer holding period.

M6 and M7 do not improve on the core network specifications. M6, which substitutes standard graph centrality features for the hypergraph block, performs below M4b and M5 for XGBoost (0.693), and only marginally above M3 for Random Forest (0.697). This indicates that the role-weighted, group-aware features motivated by the hypergraph framework carry more predictive information than classical graph-theoretic topology measures (clustering, PageRank, betweenness) computed from the pairwise projection of the same underlying syndicate data. The kitchen-sink specification M7 does not outperform M5 or M4b for any algorithm, suggesting diminishing returns from combining all feature groups and confirming that M5 is not leaving large amounts of predictive information on the table.

HGNN underperforms in this sample The underperformance of the HGNN relative to the classical models warrants comment. With 3,109 deals and a maximum training set of 2,784 observations at Fold 6, the dataset falls at the lower end of what deep learning architectures typically require to learn meaningful node embeddings from scratch. The HGNN must simultaneously learn bank representations via hypergraph convolution and a classification head, with the two-stage pretraining procedure providing only partial mitigation. Classical models, by contrast, receive the hypergraph structure as pre-computed, interpretable features and exploit it directly through XGBoost's gradient-boosted trees. In this regime, the inductive bias of hand-crafted features proves more efficient than end-to-end representation learning.

5.3 Ordinal Characterisation of the Best Specifications

The AUC-Macro results of Section 5.2 established which specifications and algorithms yield competitive discrimination under the standard evaluation metric used in the IPO prediction literature. The study now asks a more targeted question: does the model's discrimination have the ordinal structure required to support a selective investment strategy?

Standard AUC-Macro treats all pairwise ranking errors as equivalent. For an investor, however, what matters is specifically the model's ability to separate Hot deals (first-day return $> 15\%$) from Moderate ones (0–15%). I therefore extend the evaluation to a richer metric suite that explicitly penalises ordinal misclassification, restricting attention to XGBoost at d_0 under the multiclass setting. The four models examined — M2, M3, M4b, and M5 — represent, respectively, the no-network baseline, the minimal network addition, the full hypergraph specification on the core financial block, and the full hypergraph specification on the extended financial block.

5.3.1 Full Metric Suite

Table 5 reports seven metrics averaged across the six test folds. In addition to AUC-Macro, the research include: the Adjacent AUC, the average of the two boundary-level AUCs; the Hot-boundary AUC, which measures discrimination between Moderate and Hot outcomes only; Quadratic Weighted Kappa (QWK), which penalises two-step errors four times more than one-step errors; Spearman rank correlation between the model's expected class $\hat{c} = \sum_k k\,\hat{p}_k$ and the true label; and Mean Absolute Error (MAE) on the class labels.

Table 5: Ordinal and boundary metric suite for XGBoost at d_0, multiclass, averaged across six expanding-window test folds (2018–2026). ↑ = higher is better; ↓ = lower is better. Best value in each row is **bold**. M2 is the no-network baseline; M3 adds hypergraph role features; M4b adds all three hypergraph feature families on the core financial block; M5 combines extended financials with all hypergraph features. Adjacent AUC is the average of the two boundary-level AUCs; Hot-boundary AUC (1v2) measures Moderate vs Hot discrimination only; QWK penalises two-step errors quadratically; MAE is computed on class labels (0, 1, 2).

Metric	**M2**	**M3**	**M4b**	**M5**
AUC-Macro OVR ↑	0.6955	0.6923	**0.7031**	0.6980
Adjacent AUC ↑	0.6971	0.6966	**0.7068**	0.6988
Hot-boundary AUC (1v2) ↑	0.7354	0.7381	**0.7481**	0.7446
QWK ↑	0.2737	0.2696	0.2882	**0.2996**
Spearman ρ ↑	0.3408	0.3332	0.3457	**0.3481**
MAE ↓	0.6073	0.6083	0.5940	**0.5897**

Several findings emerge from Table 5.

M3 does not improve on the no-network baseline. Across all six metrics, M3 either matches or falls slightly below M2. AUC-Macro drops from 0.695 to 0.692, QWK from 0.274 to 0.270, and Spearman ρ from 0.341 to 0.333. The only metric where M3 edges ahead is the Hot-boundary AUC (0.738 vs 0.735), suggesting that the role-based hypergraph features contribute a small amount of boundary-relevant signal, but not enough to improve overall ordinal ranking. This result indicates that the minimal network addition

(HG_R) is insufficient to improve upon a well-specified no-network baseline.

M4b is the strongest specification overall. M4b achieves the highest AUC-Macro (0.703), Adjacent AUC (0.707), and Hot-boundary AUC (0.748), all above the M2 baseline by margins of +1.0, +1.0, and +1.3 percentage points respectively. The Hot-boundary AUC of 0.748 is the most economically meaningful number in the table: given an IPO that is either Moderate or Hot, the model correctly ranks the Hot deal 74.8% of the time. This is the metric that directly governs the expected precision of the selective investment strategy.

M5 leads on ordinal ranking metrics. Despite marginally lower Hot-boundary AUC than M4b (0.745 vs 0.748), M5 achieves the best QWK (0.300), Spearman ρ (0.348), and MAE (0.590). QWK of 0.300 implies substantial agreement beyond chance when ordinal distance is penalised, and the lower MAE confirms that M5 makes fewer large misclassifications. M5 is also the most stable specification, with the lowest standard deviation across folds for both QWK (0.035) and Spearman ρ (0.058), compared to M3's 0.091 and 0.079 respectively.

The Hot boundary is substantially easier to predict than the loss boundary. Across all four specifications, Hot-boundary AUC (Moderate vs Hot) exceeds the Loss boundary AUC (Negative vs Moderate) by approximately 8 percentage points (0.735–0.748 vs 0.653–0.666). This asymmetry reflects the fact that Hot IPOs tend to exhibit stronger pre-issue signals — larger deal sizes, more prestigious syndicates, more favourable market conditions — that the model can separate from Moderate outcomes with higher confidence. The loss boundary, by contrast, is harder to predict: Negative outcomes are more idiosyncratic and less systematically related to pre-issue observables. For the trading strategy, this asymmetry is reassuring: the model is most accurate precisely at the boundary that matters.

5.3.2 Fold-by-Fold Stability

Figure 2 plots all four metrics from Table 5 across the six test folds for each model.

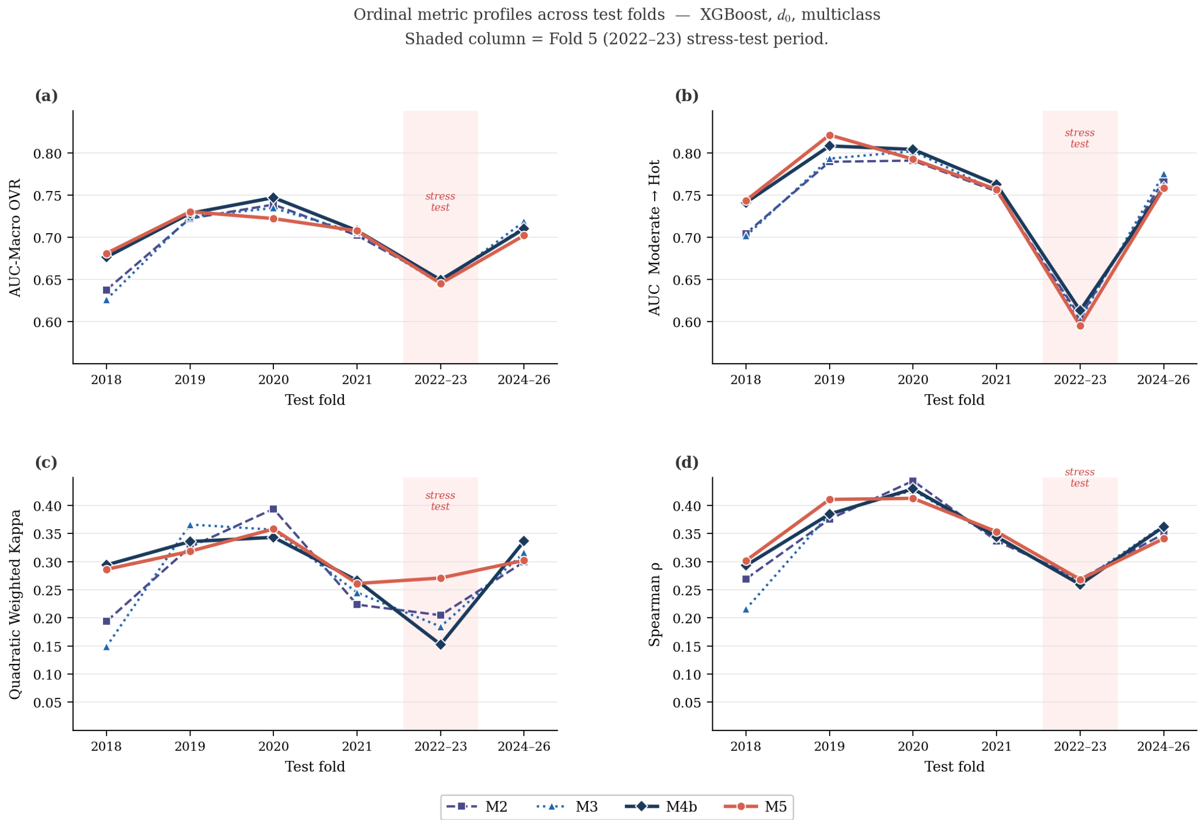

Figure 2: Ordinal metric profiles across test folds for XGBoost at d_0, multiclass. Each line traces one model specification across the six expanding-window test folds (2018–2026). M2: no-network baseline; M3: adds hypergraph role features; M4b: full hypergraph feature set on core financials; M5: full hypergraph set on extended financials. The shaded column marks Fold 5 (2022–23), a stress period of compressed IPO volumes under Federal Reserve tightening. Panels: (a) AUC-Macro OVR, one-vs-rest macro-averaged; (b) Hot-boundary AUC, measuring Moderate vs Hot discrimination only; (c) Quadratic Weighted Kappa, penalising ordinal misclassification distance; (d) Spearman ρ, rank correlation between the model's expected class index and the true label.

Performance peaks in 2019 (Fold 2) and 2020 (Fold 3) across all metrics and all models. AUC-Macro reaches 0.730 for M5, QWK reaches 0.394 for M2 in 2020, and Hot-boundary AUC exceeds 0.80 in both years. These were high-volume periods with wide underpricing dispersion: the 2019 technology cycle and the 2020 post-COVID reopening wave both generated conditions in which syndicate composition carried strong predictive signal. The model learns best when the market provides heterogeneity to separate.

Conversely, every metric and every model deteriorates in 2022–23. AUC-Macro falls to 0.645–0.649, Hot-boundary AUC collapses to 0.595–0.613, and QWK drops below 0.20

for M3 and M4b. This fold covers the Federal Reserve tightening cycle: sharply rising interest rates compressed IPO volumes to near-decade lows (only 254 test deals, the smallest fold), and the underpricing patterns that characterise normal market conditions became unreliable. Notably, M5 is the most resilient specification in this fold on the ordinal metric: it maintains a QWK of 0.271, substantially above M3 (0.185) and M4b (0.153). On Hot-boundary AUC the ordering reverses (M5: 0.595, M3: 0.602, M4b: 0.613), consistent with M5's advantage residing in global ordinal calibration rather than boundary discrimination under low-volume stress conditions.

In Folds 1, 2, 3, and 4, M4b achieves the highest Hot-boundary AUC in three of four cases, and M5 leads in QWK and Spearman ρ in three of four. M3 and M2 are competitive in some folds — M3 slightly outperforms in Fold 6 on AUC-Macro (0.719 vs 0.711 for M4b) — but show no systematic advantage. The consistency of M4b and M5 above the no-network baseline M2 in all but the stress-test fold supports the conclusion that hypergraph network features contribute genuine predictive signal, with the caveat that this signal is partly attenuated in low-volume, high-stress market regimes.

5.4 Summary and Link to the Trading Strategy

The classification results establish three conclusions. First, pre-IPO observable characteristics — deal structure, firm fundamentals, and macroeconomic conditions — contain genuine out-of-sample predictive information about first-day return outcomes. The dominant performance gain occurs before any network information enters: M0 to M2 accounts for the majority of discriminative lift across all algorithms, confirming that firm fundamentals and market conditions are the primary drivers of IPO underpricing predictability.

Second, hypergraph network features improve discrimination for XGBoost, with M4b and M5 consistently outperforming the no-network baseline M2 outside the 2022–23 stress period. This improvement is algorithm-specific: Random Forest and Logistic Regression do not benefit from network augmentation, indicating that the network signal requires a model capable of learning feature interactions to be extracted. The role-weighted, group-aware features motivated by the hypergraph framework outperform classical graph-theoretic topology measures, as evidenced by M6 underperforming M4b and M5.

Third, the HGNN does not outperform the classical models despite consuming the same underlying relational data. This can be attributed to sample size constraints: with fewer than 3,000 deals, end-to-end representation learning does not recover the efficiency of hand-crafted hypergraph features fed to a gradient-boosted ensemble.

Among the network-augmented specifications, M5 is selected as the primary model for the trading strategy. The choice reflects a trade-off between boundary discrimination and global ordinal calibration. M4b achieves marginally higher Hot-boundary AUC, indicating sharper pairwise separation of Moderate from Hot outcomes; M5 achieves superior QWK and MAE, indicating fewer severe misclassifications. For a threshold-based strategy that invests in all deals above a $\hat{p}(\text{Hot})$ cutoff, the latter property is more consequential: a Moderate deal entering the portfolio dampens returns, but a Negative deal destroys them. The fold-by-fold analysis of Section 5.3.2 reinforces this choice: M5 is the most stable specification across test periods, maintaining the highest QWK even in the 2022–23 stress fold where boundary-level metrics deteriorate for all models. Finally, as the saturated hypergraph specification, M5 provides the most complete test of whether syndicate network features improve IPO return prediction. M2 and M3 serve as baselines — the no-network reference and the minimal network addition respectively — and the $\hat{p}(\text{Hot})$ probability from the multiclass model at d_0 serves directly as the investment signal.

6 Trading Strategy Evaluation

This section translates the classification results of Section 5 into an economic question: does the pre-IPO signal generated by the XGBoost models support a profitable selective subscription strategy? I design a simple rule-based strategy, evaluate it across the same six expanding-window test folds, and compare outcomes against a passive subscriber and a simulated random-selection benchmark.

6.1 Strategy Design and Assumptions

An IPO investor must decide whether to request an allocation before trading begins, as the first-day return accrues entirely between the offer price and the first-day close. The strategy is therefore binary: subscribe or pass. The investor allocates money in deal i if the multiclass model assigns $\hat{p}_i(\text{Hot}) \geq \tau$ for a threshold τ, and weight all selected deals equally. Within each test fold, the mean first-day return of the selected portfolio is computed; cumulative performance is then obtained by compounding these six fold-level returns sequentially, so that the final portfolio value reflects the multiplicative growth of a \$1.00 investment reinvested across the six non-overlapping test periods. The outcome variable is the offer-to-first-day-close return, used as a proxy for the underpricing pop accruing to allocation holders.

All features are derived from data available before the listing date, and training never uses information from the test period. I assume full allocation access, which inflates returns relative to live performance in oversubscribed deals. Additionally, bookrunners informally penalise early selling by reducing future allocation access, so the first-day return is not literally realisable through immediate selling — it should be read as measuring the economic value of the allocation, not as a deployable trading rule. M2 (no network), M3 (partial hypergraph), and M5 (full hypergraph) are tested across a range of thresholds $\tau \in [0.33, 0.70]$.

Two benchmarks provide the evaluation context. The passive subscriber invests in every IPO with equal weight without forecasting ability. The random selection benchmark draws 10,000 portfolios without replacement from the test universe per fold, each matched to M5's selection size, and compounds the results across folds. The min–max band of this simulation provides a stringent test of whether model skill rather than portfolio concentration drives performance.

6.2 Threshold Sweep and Model Comparison

Figure 3 shows the cumulative return for M2, M3, and M5 across all tested thresholds.

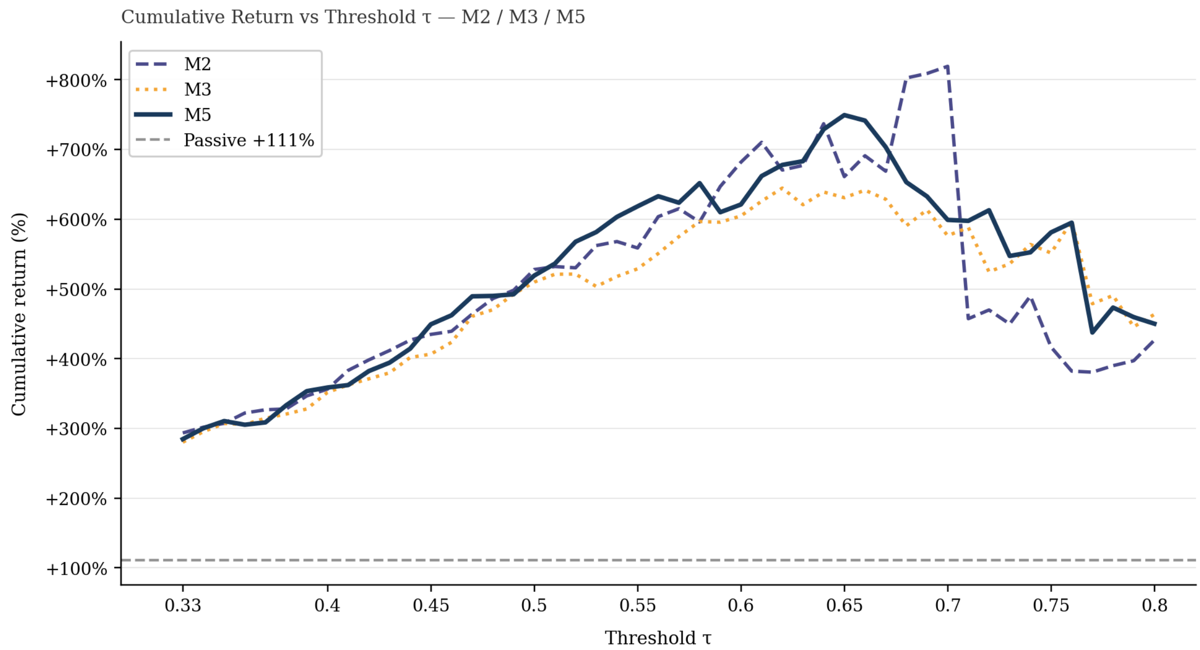

Figure 3: Portfolio cumulative return for M2, M3, and M5 across thresholds $\tau \in [0.33, 0.80]$, multiclass d_0. The strategy subscribes to all IPOs with $\hat{p}(\text{Hot}) \geq \tau$ and weights selected deals equally; cumulative value compounds six fold-level returns (2018–2026). M2: no-network baseline; M3: adds hypergraph role features; M5: extended financials with all hypergraph features. The dashed grey line shows the passive subscriber (+111%), who invests in every IPO without selection. No pairwise model difference is statistically significant at any threshold (paired t-test on fold-level returns, all $p > 0.19$, $n = 6$; deal-level McNemar test, $p = 0.20$; see Section 6.2). Beyond $\tau \approx 0.60$, portfolio size falls below 50 deals per fold and results become volatile.

First, all three models improve with τ through the range $\tau \leq 0.60$, confirming that $\hat{p}(\text{Hot})$ is a monotonically informative ranking signal: deals assigned higher Hot probability generate higher average first-day returns on average regardless of which model produces the signal.

Second, M2, M3, and M5 are not statistically distinguishable at any threshold. A paired t-test on fold-level returns finds no significant difference between any pair of models across the entire tested range (all p-values > 0.19, $n = 6$ folds). To confirm that this is not an artefact of the low-powered fold-level comparison, I conduct two deal-level tests on the 2,092 individual IPO predictions pooled across all six test folds. A McNemar test on classification correctness finds that M5 is correct on 159 deals where M2 is wrong and wrong on 136 where M2 is correct, a directional advantage that does not reach significance ($\chi^2 = 1.64$, $p = 0.20$). A disagreement analysis comparing the realised returns of deals selected exclusively by one model yields a similar conclusion: at $\tau = 0.55$, M5's unique selections average 36.4% with a 68.4% hit rate[4] against 25.6% and 54.7% for M2's, but the gap does not reach conventional significance (Mann–Whitney $p = 0.067$) and vanishes at lower thresholds.[5] Therefore, the study cannot claim that network features improve trading performance over the no-network baseline.

Third, M3 is the weakest specification across the range, falling below both M2 and M5 at five of six discrete thresholds in the stable regime. This is consistent with the ordinal metric finding in Section 5.3.1: partial hypergraph augmentation without the full structural representation degrades probability calibration rather than improving it, and this degradation persists at the trading level.

Beyond $\tau \approx 0.60$, portfolio size falls below 50 deals per fold and results become volatile and unreliable, with final values swinging by more than $2.00 between adjacent thresholds for all models. Results are not interpreted in this regime.

6.3 Performance Against Passive and Random Benchmarks

Although the models cannot be distinguished from each other, I test whether the strategy as a whole generates economically meaningful returns. Table 6 reports results at three representative thresholds for M5.

Table 6: M5 multiclass d_0 strategy at three thresholds versus passive and random benchmarks. M5 is the saturated specification combining extended financials with all hypergraph features. The strategy subscribes to all IPOs with $\hat{p}(\text{Hot}) \geq \tau$ and weights selected deals equally. Cumulative value compounds six fold-level mean returns (2018–2026) from a $1.00 base. Hit rate = fraction of selected deals with $r_i > 15\%$. The passive subscriber invests in every IPO without selection. The Random column reports the mean across 10,000 size-matched random portfolios (averaged across the three thresholds). The absolute maximum across all simulations was $3.89 (+289%); M5 exceeds every single random portfolio at all three thresholds.

	M5 strategy				
	$\tau = 0.45$	$\tau = 0.50$	$\tau = 0.55$	**Passive**	**Random**
Final $1.00 invested	$5.49	$6.19	$7.18	$2.11	$2.11
Total return	+449%	+519%	+618%	+111%	+111%
Mean fold return	33.2%	35.9%	39.5%	13.4%	13.4%
Avg deals per fold	83.7	69.3	58.2	416.0	70.4
Hit rate ($r > 15\%$)	65.3%	69.8%	74.5%	34.1%	31.7%

[4] Hit rate is the percentage of selected deals with $r_i > 15\%$

[5] A per-fold McNemar decomposition shows M5 favoured in four of six folds with no individual fold reaching significance, indicating the directional advantage is diffuse rather than driven by a single test period.

At every tested threshold M5 generates cumulative returns between +449% and +618%, against a passive benchmark of +111% and a random simulation maximum of +289% across 10,000 draws. M5 exceeds every single random portfolio at all three thresholds shown.

Figure 4 shows the cumulative portfolio evolution for M5 at the three thresholds alongside the passive benchmark and the random min–max band. The separation from both benchmarks is consistent from the first fold and widens over time.

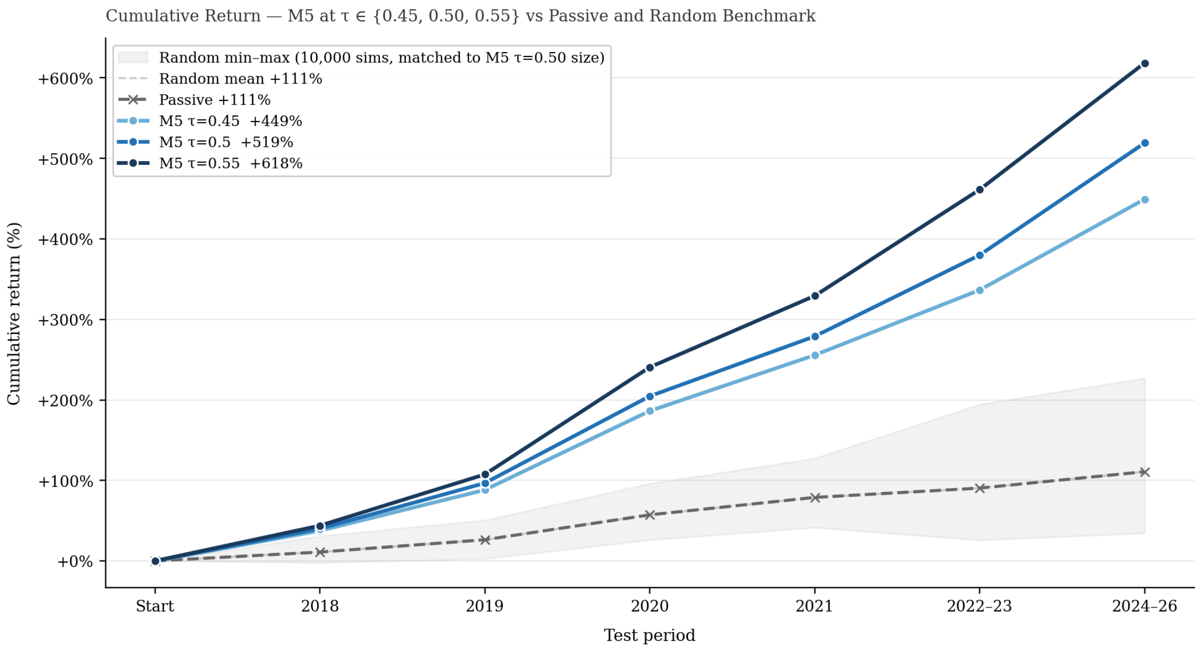

Figure 4: Cumulative portfolio evolution for M5 at $\tau \in \{0.45, 0.50, 0.55\}$ versus the passive benchmark and the min–max band of 10,000 random portfolios, across the six expanding-window test folds (2018–2026). M5 is the saturated specification combining extended financials with all hypergraph features. The strategy subscribes to all IPOs with predicted Hot probability $\hat{p}(\text{Hot}) \geq \tau$ and weights selected deals equally; each step compounds the fold-level mean return.The passive subscriber (dashed grey) invests in every IPO without selection. The shaded band spans the minimum and maximum of 10,000 random portfolios drawn without replacement from the test universe per fold, matched to M5's selection size at $\tau = 0.50$. M5 separates from both benchmarks consistently from the first fold and exceeds every random portfolio at all three thresholds.

Table 7 decomposes the M5 strategy at $\tau = 0.50$ by test fold. Every fold is profitable, including Fold 5 (2022–23), which represents the stress period identified in Section 5.3.2: a sharp contraction in IPO volume under Federal Reserve tightening, with only 13 deals selected and the lowest fold return of 26.6%. The strategy degrades but does not fail under stress. Figure 5 shows the per-fold returns for all three models at $\tau = 0.50$ against the passive benchmark. The consistency of the outperformance across folds and models is evident proving that results are not driven by a single period.

Table 7: Per-fold returns for M5 multiclass at $\tau = 0.50$ versus the passive subscriber, across the six expanding-window test folds (2018–2026). M5 is the saturated specification combining extended financials with all hypergraph features; the strategy subscribes to all IPOs with $\hat{p}(\text{Hot}) \geq 0.50$ and weights selected deals equally. Lift = M5 return minus passive benchmark return. n = number of deals selected by M5 in each fold. The passive subscriber invests in every IPO without selection. Every fold is profitable, including Fold 5 (2022–23), a stress period of compressed IPO volumes under Federal Reserve tightening.

Test period	**M5 return**	**Passive**	**Lift (%)**	n
2018	40.1%	11.0%	+29.1	32
2019	40.4%	13.8%	+26.6	45
2020	54.9%	24.5%	+30.4	73
2021	24.4%	13.8%	+10.7	208
2022–23	26.6%	6.5%	+20.2	13
2024–26	29.0%	10.7%	+18.3	45
Mean	**35.9%**	**13.4%**	**+22.5pp**	**69.3**

6.4 Interpretation

The trading evaluation yields two findings of differing strength.

Firstly, a selective IPO subscription strategy based on $\hat{p}(\text{Hot})$ generates large, consistent returns that completely dominate passive subscription and random selection at every reasonable threshold. With six out-of-sample test periods spanning diverse market conditions — including a stress period in 2022–23 and the 2020 underpricing boom — the result is not an artifact of a single favourable year. The pre-IPO feature set captures genuine information about first-day return outcomes.

The second finding is a null result: the addition of hypergraph network features does not produce a statistically distinguishable improvement in simulated trading performance. M5 and M2 generate statistically equivalent returns at every tested threshold, both at the fold level and at the deal level. This does not contradict the classification results of Chapter 5, where M5's superior Hot-boundary AUC and ordinal metrics are well-established. Deal-level tests on the 2,092 individual predictions confirm that the improvement, while directionally consistent, is too small in magnitude to reach statistical significance even with full per-IPO resolution (Section 6.2). The practical implication is straightforward: the trading strategy's performance is robust to model choice. Network features improve

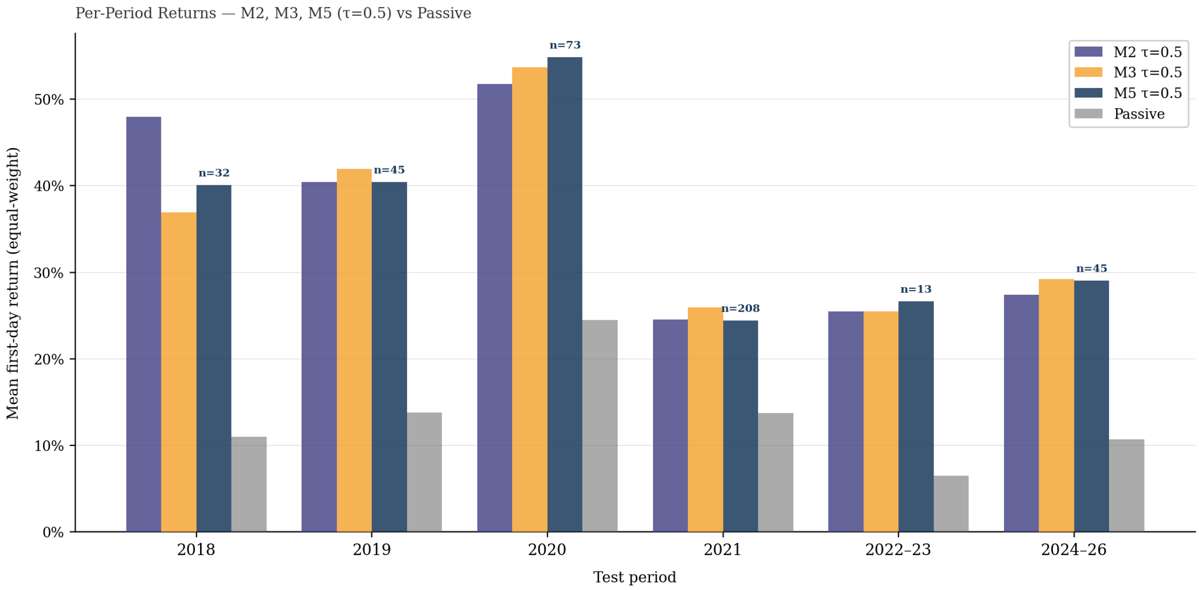

Figure 5: Per-fold mean first-day returns for M2, M3, and M5 at $\tau = 0.50$ versus the passive benchmark, across the six expanding-window test folds (2018–2026). The strategy subscribes to all IPOs with $\hat{p}(\text{Hot}) \geq 0.50$ and weights selected deals equally. M2: no-network baseline (market controls, deal characteristics, core financials); M3: adds hypergraph role features; M5: combines extended financials with all hypergraph features. The passive subscriber (grey) invests in every IPO without selection. Annotations above M5 bars show the number of deals selected per fold. All three models remain profitable in every fold, including the 2022–23 stress period (Fold 5, 13 deals selected).

the classification metrics, as shown in Chapter 5, but at the portfolio level all three specifications perform alike.

7 Conclusions

The study investigates whether the structure of underwriting syndicates, represented as an evolving hypergraph, carries predictive information about IPO first-day returns. Using 3,109 deals from the US, Canada, and Europe between 2013 and 2026, it compares hand-crafted hypergraph features against standard pairwise graph features and against a Hypergraph Neural Network.

Evidence shows that traditional IPO pricing determinants (financials, deal characteristics and macro-level variables) remain the primary drivers of underpricing predictability. However, hypergraph information moderately improves predictive performance in multiclass classification for XGBoost, with the full hypergraph specifications outperforming both the no-network baseline and the standard pairwise graph specification on macro-AUC and ordinal metrics. Random Forest and Logistic Regression do not benefit from network augmentation, indicating that the syndicate signal requires a model capable of learning feature interactions to be extracted. Moreover, the HGNN does not match the performance of hand-crafted features fed to gradient-boosted trees and this result is mainly attributed to sample size constraints on end-to-end representation learning. Finally, when translated into a selective trading strategy, the pre-IPO signal generates cumulative returns that dominate passive subscription and random benchmarks across all market regimes, though the incremental contribution of network features is not statistically distinguishable at the portfolio level.

Several limitations qualify these conclusions. The sample, while sufficient for tabular machine learning, is small by deep learning standards and limits both the viability of the HGNN and the statistical power of model comparisons at the trading level. The trading strategy assumes full allocation access and does not account for the partial allocation and implicit selling constraints that characterise real IPO markets. Financial data coverage for pre-IPO firms is incomplete, with missing value rates exceeding 25% for some variables, and the role-weighting scheme, while ordinally motivated, has not been formally optimised.

These limitations set the ground for future research. A larger sample would enable a more definitive evaluation of the HGNN and stronger statistical tests of the network features' trading value. Richer pre-IPO financial data, including granular accounting items and textual features from prospectus filings, could improve the baseline prediction and clarify the marginal contribution of syndicate structure. Finally, the current strategy evaluates only the first-day return, which is the relevant decision point for allocation but does not exploit the model's predictions at longer horizons. A more realistic evaluation would incorporate a buy-and-hold or active trading rule over the d_9, d_{20} and other further horizons, allowing the strategy to capture post-listing drift and to manage positions dynamically based on evolving probability estimates.

References

Akiba, Takuya et al. (2019). *Optuna: A Next-generation Hyperparameter Optimization Framework*. arXiv: 1907.10902 [cs.LG]. URL: https://arxiv.org/abs/1907.10902.

Alahmadi, Mazin (2025). "A Risk-Optimized Framework for Data-Driven IPO Underperformance Prediction in Complex Financial Systems". In: *Systems* 13.3, p. 179. DOI: 10.3390/systems13030179.

Allen, Franklin and Gerald R. Faulhaber (1989). "Signalling by underpricing in the IPO market". In: *Journal of Financial Economics* 23.2, pp. 303–323. DOI: None. URL: https://ideas.repec.org/a/eee/jfinec/v23y1989i2p303-323.html.

Bajo, Emanuele et al. (2016). "Underwriter networks, investor attention, and initial public offerings". In: *Journal of Financial Economics* 122.2, pp. 376–408. ISSN: 0304405X. DOI: 10.1016/j.jfineco.2015.12.001. URL: https://linkinghub.elsevier.com/retrieve/pii/S0304405X1630126X (visited on 03/01/2026).

Battiston, Federico et al. (2020). "Networks beyond pairwise interactions: Structure and dynamics". In: *Physics Reports* 874. DOI: 10.1016/j.physrep.2020.05.004.

Benveniste, Larry and Paul Spindt (1989). "How Investment Bankers Determine the Offer Price and Allocation of New Issues". In: *Journal of Financial Economics* 24, pp. 343–361. DOI: 10.1016/0304-405X(89)90051-2.

Berge, Claude and Edward Minieka (1973). *Graphs and Hypergraphs*. Amsterdam: North Holland Publishing Company.

Booth, James R. and Richard L. Smith (1986). "Capital raising, underwriting and the certification hypothesis". In: *Journal of Financial Economics* 15, pp. 261–281. URL: https://api.semanticscholar.org/CorpusID:153610830.

Bradley, Daniel J and Bradford D Jordan (2002). "Partial adjustment to public information and IPO underpricing". In: *Journal of Financial and Quantitative Analysis* 37.4, pp. 595–616.

Breiman, Leo (2001). "Random forests". In: *Machine learning* 45.1, pp. 5–32.

Bretto, Alain (2013). *Hypergraph Theory: An Introduction*. Cham: Springer. DOI: 10.1007/978-3-319-00080-0.

Carter, Richard and Steven Manaster (1990). "Initial Public Offerings and Underwriter Reputation". In: *The Journal of Finance* 45.4, pp. 1045–1067. ISSN: 00221082, 15406261. URL: http://www.jstor.org/stable/2328714 (visited on 03/06/2026).

Chemmanur, Thomas J. and Karthik Krishnan (2012). "Heterogeneous Beliefs, Short Sale Constraints, and the Economic Role of the Underwriter in IPOs". In: *Financial Management* 41.4, pp. 769–811. DOI: 10.2139/ssrn.890859.

Chen, Tianqi and Carlos Guestrin (2016). "XGBoost: A Scalable Tree Boosting System". In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. DOI: 10.1145/2939672.2939785. URL: http://dx.doi.org/10.1145/2939672.2939785.

Chuluun, Tuugi (2015). "The role of underwriter peer networks in IPOs". In: *Journal of Banking & Finance* 51, pp. 62–78. ISSN: 03784266. DOI: 10.1016/j.jbankfin.2014.11.001. URL: https://linkinghub.elsevier.com/retrieve/pii/S037842661400346X (visited on 03/01/2026).

Corwin, Shane A. and Paul Schultz (2005). "The Role of IPO Underwriting Syndicates: Pricing, Information Production, and Underwriter Competition". In: *Journal of Finance* 60.1, pp. 443–486. DOI: 10.1111/j.1540-6261.2005.00735.x.

FactSet Research Systems Inc. (2026). *FactSet Deals Screening: Initial Public Offerings*. Proprietary financial database. IPO transaction-level data including syndicate roles and underwriting information.

Feng, Yifan et al. (2019). *Hypergraph Neural Networks*. arXiv: 1809.09401 [cs.LG]. URL: https://arxiv.org/abs/1809.09401.

Jain, Bharat A. and Omesh Kini (1994). "The Post-Issue Operating Performance of IPO Firms". In: *The Journal of Finance* 49.5, pp. 1699–1726. ISSN: 00221082, 15406261. URL: http://www.jstor.org/stable/2329268 (visited on 03/11/2026).

Ljungqvist, Alexander and William J. Wilhelm (2003). "IPO Pricing in the Dot-com Bubble". In: *Journal of Finance* 58.2, pp. 723–752. URL: https://EconPapers.repec.org/RePEc:bla:jfinan:v:58:y:2003:i:2:p:723-752.

Loughran, Tim and Bill McDonald (2013). "IPO first-day returns, offer price revisions, volatility, and form S-1 language". In: *Journal of Financial Economics* 109.2, pp. 307–326. URL: https://EconPapers.repec.org/RePEc:eee:jfinec:v:109:y:2013:i:2:p:307-326.

Megginson, William L and Kathleen A Weiss (1991). "Venture Capitalist Certification in Initial Public Offerings". In: *Journal of Finance* 46.3, pp. 879–903. URL: https://EconPapers.repec.org/RePEc:bla:jfinan:v:46:y:1991:i:3:p:879-903.

Ritter, Jay and Tim Loughran (2004). "Why Has Underpricing Changed Over Time?" In: *Financial Management* 33. DOI: 10.2139/ssrn.331780.

Ritter, Jay R. (1984). "The "Hot Issue" Market of 1980". In: *The Journal of Business* 57.2, pp. 215–240. (Visited on 03/04/2026).

Ritter, Jay R. and Ivo Welch (2002). "A Review of IPO Activity, Pricing, and Allocations". In: *Journal of Finance* 57.4, pp. 1795–1828. DOI: 10.1111/1540-6261.00478.

Rock, Kevin F. (1986). "Why new issues are underpriced". In: *Journal of Financial Economics* 15, pp. 187–212. URL: https://api.semanticscholar.org/CorpusID:18680572.

Tsui, Christophor Sai Kit and Kevin Chi Keung Li (2021). "IPO Underpricing and Prospectus Readability: A Machine Learning Approach". In: *Journal of Applied Business and Economics* 23.7, pp. 151–166.

Welch (1989). "Seasoned Offerings, Imitation Costs, and the Underpricing of Initial Public Offerings". In: *The Journal of Finance* 44.2, pp. 421–449. DOI: https://doi.org/10.1111/j.1540-6261.1989.tb05064.x.

Wilhelm, William and Pegaret Pichler (2001). *A Theory of the Syndicate: Form Follows Function*. Economics Series Working Papers 2001-FE-05. University of Oxford, Department of Economics. URL: https://EconPapers.repec.org/RePEc:oxf:wpaper:2001-fe-05.

AI Acknowledgment

During the preparation of this thesis, I made use of generative artificial intelligence tools. In particular, Claude was used to refine the English language to improve the clarity and readability of selected passages. In addition, I used Claude and ChatGPT to assist with minor revisions and debugging of python code used in the empirical analysis.

These tools were employed solely as support instruments. All substantive research design choices, data construction, econometric specifications, interpretation of results, and conclusions are my own. I carefully reviewed and edited all AI-assisted outputs and take full responsibility for the final content of this thesis.

A Hypergraph and Graph Feature Definitions

Hypergraph Role Features (HGR, 5 features + 1 in M7 only) These features capture the internal composition of the syndicate, measuring intensity and diversity.

- `hg_role_sum` Total role weight across syndicate members; a proxy for syndicate capacity.
- `hg_role_mean` Average role weight per bank; captures the average seniority of participants.
- `hg_role_max` Weight of the most senior bank; measures leadership strength.
- `hg_role_min` Weight of the most junior bank; captures the depth of the syndicate hierarchy.
- `hg_role_entropy`. Shannon entropy of the role share distribution. It ranges from 0 (a single bank holds all weight) to log(num of participating banks) (all banks hold equal weight). Higher entropy indicates a more evenly distributed syndicate.
- `hg_role_hhi` (*M7 only*). Herfindahl–Hirschman Index of role share concentration. Measures the same concept as entropy but from the opposite direction. Their em-

pirical correlation in this sample is $r = 0.93$. HHI is therefore excluded from the primary specifications and included only in the kitchen-sink specification M7.

Bank-level activity features (HGP, 4 features) These features measure the historical activity and experience of the banks participating in the focal deal, computed from the rolling three-year window of prior syndicate activity.

- `hg_sum_bank_wdeg` Total weighted degree across all syndicate members; captures the aggregate historical activity of the syndicate.
- `hg_mean_bank_wdeg` Mean weighted degree per bank; measures the average experience level of participants.
- `hg_max_bank_wdeg` Weighted degree of the most active bank in the syndicate; captures whether the deal includes a highly experienced institution.
- `hg_num_banks_in_snapshot`. Total number of distinct banks active in at least one deal within $\mathcal{W}_i$. This controls for market-wide syndicate density at the time of the deal: in active markets (e.g. 2020–21) many banks participate, while in cold markets (e.g. 2022–23) fewer are active. Including this variable prevents the model from confusing an experienced syndicate with a busy market.

Additional Hypergraph and Pairwise Features (HGA, 5 features) This family extends the network characterisation to pairwise collaboration history and leadership structure within the syndicate.

- `hg_pair_co_count_mean`. Mean pairwise co-underwriting count across all syndicate pairs; captures the average depth of prior collaboration among participants.
- `hg_pair_co_count_max` Maximum pairwise co-count; identifies the strongest existing relationship in the syndicate.
- `hg_pair_co_frac_pos`. Fraction of syndicate pairs with at least one prior co-deal. This measures how familiar the syndicate is as a working unit — whether its members have established relationships or form an ad hoc assembly.

- `hg_lead_wdeg`. Weighted degree of the lead bank (the bank with the highest role weight in deal i), computed from the rolling window as defined in $\mathrm{HG_P}$. Isolates the historical experience of the most senior syndicate member specifically.

- `hg_dominance_ratio`. Ratio of the lead bank's role weight to the total syndicate role weight. Higher values indicate a more top-heavy syndicate in which the lead bank concentrates a disproportionate share of seniority.

Standard Graph Features (GR, 8 features) As a benchmark for the hypergraph representation, a parallel set of features is constructed from the standard pairwise co-underwriting graph $G_i = (V_i, E_i)$, in which two banks are connected by an edge whenever they have co-underwritten at least one deal within the rolling window. Unlike the hypergraph-derived features above, these measures exploit graph-theoretic properties that operate on binary pairwise topology, without access to role weights or group-level syndicate structure.

- `g_clustering_mean`, `g_clustering_max`. Mean and maximum local clustering coefficient across syndicate members. The clustering coefficient of bank b measures the fraction of b's neighbours that are themselves connected. High clustering indicates that a bank operates within a tightly knit group of co-underwriters.

- `g_pagerank_mean`, `g_pagerank_max`. Mean and maximum PageRank centrality across syndicate members. PageRank assigns higher scores to banks that are connected to other highly connected banks, capturing prestige through association in the co-underwriting network.

- `g_betweenness_mean`, `g_betweenness_max`. Mean and maximum betweenness centrality across syndicate members. Betweenness measures the fraction of shortest paths in the network that pass through bank b; high values indicate that the bank serves as a bridge between otherwise disconnected parts of the co-underwriting network.

- `g_snapshot_n_edges`. Total number of edges in the snapshot graph G_i; controls for overall market-wide co-underwriting activity at the time of the deal.
- `g_snapshot_density` Fraction of possible pairwise connections that exist in the snapshot; captures how interconnected the co-underwriting market is at the time of the deal.

These features enter only M6 and M7. Their inclusion in M6 — where they substitute for the hypergraph blocks — provides a direct test of whether graph-theoretic centrality measures capture the same information as the role-weighted, group-aware features derived from the hypergraph.

Role-weighting scheme

For each deal i, bank b is assigned the maximum weight $w_{i,b}$ corresponding to its highest role in that deal. The *role share* of bank b is then $s_{i,b} = w_{i,b} / \sum_{b'} w_{i,b'}$.

Table 8: Role weight assignments. Each bank is assigned the weight corresponding to its highest role in a given deal.

Syndicate role	**Weight**
Global Coordinator	4.0
Joint Global Coordinator	3.5
Bookrunner	3.5
Joint Bookrunner	3.0
Lead Manager	2.5
Co-Lead Manager	2.0
Co-Manager	1.5
Broker	1.0
Nominated Advisor	1.0
Selling Group Member	0.5

B HGNN Training Procedure

First, a 100-epoch self-supervised pre-training warm-up optimises the HGNN encoder on a masked hyperedge reconstruction objective: 15% of hyperedges are randomly masked and the encoder is trained to recover the binary co-participation structure, encouraging representations that reflect syndicate co-occurrence before any return signal is introduced. Second, the full model — encoder and MLP head jointly — is fine-tuned on the classification task for up to 400 epochs using cross-entropy loss with inverse-frequency class weights. The encoder and head use separate learning rates (5×10^{-5} and 3×10^{-4} respectively) to prevent the stronger gradient signal from the head destabilising the pre-trained encoder. A `ReduceLROnPlateau` scheduler and early stopping with patience 100 are applied throughout fine-tuning. Hyperparameters are tuned via Optuna (Akiba et al. 2019) with 30 trials on Fold 3 and held fixed across all folds.

C Full Results

Tables 9–11 report the complete classification results across all algorithms, specifications, and horizons. All values are means across six expanding-window test folds.

Table 9: Multiclass AUC-Macro OVR at d_0, all algorithms and specifications. Values are means across six expanding-window test folds. HGNN does not consume hand-crafted feature tiers.

Spec.	XGBoost	Random Forest	Logistic Reg.	HGNN
M0	0.6110	0.6088	0.6301	—
M1	0.6734	0.6685	0.6610	—
M2	0.6955	**0.7003**	0.6681	—
M3	0.6923	0.6924	0.6676	—
M4	0.6953	0.6925	0.6674	—
M4b	**0.7031**	0.6956	0.6678	—
M5	0.6980	0.6930	**0.6726**	—
M6	0.6930	0.6970	0.6670	—
M7	0.6964	0.6928	0.6723	—
HGNN	—	—	—	**0.6135**

Bold = best value per column. — = not applicable.

Table 10: Binary AUC at d_0, all algorithms and specifications. Values are means across six expanding-window test folds.

Spec.	XGBoost	Random Forest	Logistic Reg.	HGNN
M0	0.5089	0.5269	0.5744	—
M1	0.6152	0.6016	0.6121	—
M2	0.6517	0.6557	0.6237	—
M3	0.6578	0.6522	0.6264	—
M4	0.6591	0.6438	0.6269	—
M4b	0.6530	0.6594	0.6289	—
M5	**0.6662**	**0.6657**	**0.6406**	—
M6	0.6489	0.6457	0.6228	—
M7	0.6600	0.6502	0.6403	—
HGNN	—	—	—	**0.5942**

Bold = best value per column. — = not applicable.

Table 11: Multiclass AUC-Macro OVR by return horizon for XGBoost and HGNN, all specifications. Values are means across six expanding-window test folds.

Spec.	d_0	d_9	d_{20}
M0	0.6110	0.5892	0.5909
M1	0.6734	0.6618	0.6573
M2	0.6955	0.6914	0.6789
M3	0.6923	**0.6929**	**0.6839**
M4	0.6953	0.6916	0.6830
M4b	**0.7031**	0.6893	0.6812
M5	0.6980	0.6891	0.6785
M6	0.6930	0.6897	0.6802
M7	0.6964	0.6872	0.6803
HGNN	0.6135	0.6073	0.6065

Bold = best value per column. HGNN excluded from bolding as it operates on a different feature paradigm.